

Acoustic-Levitated Dynamic Voxel System for Volumetric Field Visualization

A Physical Application of Single Entity Field Interpretation (SEFI) for Acoustic and Spatial Audio Systems

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Abstract

This paper presents a dynamic volumetric rendering system that uses acoustic levitation to position and manipulate a single illuminated particle as a free-space voxel. By controlling the particle’s trajectory through ultrasonic phased-array modulation, the system produces continuous three-dimensional curves and structures via persistence of vision. A geometric field model—specifically the Single Entity Field Interpretation (SEFI)—serves as the trajectory generator, mapping worldlines, curvature fields, and layered field constructs into real-time voxel motion[cite: 2]. Illumination synchronized with particle position encodes additional geometric or physical attributes, enabling expressive volumetric representations within an open interaction volume[cite: 2].

The proposed architecture is low-cost, safe, and based entirely on established acoustic levitation techniques[cite: 2]. It integrates a phased-array controller, a SEFI-driven coordinate engine, and a synchronized illumination subsystem to create a compact, research-grade instrument for field visualization[cite: 2]. This approach offers a direct physical embodiment of geometric computation[cite: 2]. The resulting platform supports scientific visualization, mixed-reality interaction, and experimental field modeling, providing a practical foundation for future volumetric display research and computational geometry exploration[cite: 2].

1 Introduction

Volumetric displays aim to create perceivable three-dimensional structures in physical space, enabling users to observe geometry, motion, and field behavior without reliance on flat screens or head-mounted devices[cite: 2]. Existing approaches are often constrained by limited viewing angles, high system complexity, environmental sensitivity, or significant operational cost[cite: 2]. These limitations motivate the development of a simpler, safer, and more direct method for generating dynamic volumet-

ric content[cite: 2].

Acoustic levitation provides a practical foundation for free-space voxel generation[cite: 2]. Ultrasonic phased arrays can create stable pressure nodes capable of suspending small particles at precise coordinates within an open volume[cite: 2]. By modulating the phase relationships across the array, the levitated particle can be repositioned rapidly and smoothly, enabling dynamic trajectories through three-dimensional space[cite: 2]. When illuminated in sync with its motion, the particle acts as a controllable voxel whose path forms volumetric curves and structures through persistence of vision[cite: 2].

This paper introduces a volumetric rendering system that uses a single acoustically levitated particle as a dynamic voxel driven by a geometric field model[cite: 2]. The Single Entity Field Interpretation (SEFI) provides the mathematical framework for generating worldlines, curvature fields, and layered field constructs[cite: 2]. These geometric outputs are translated into real-time voxel trajectories, allowing the levitated particle to physically trace the structure of the underlying field model[cite: 2]. Illumination synchronized with particle position encodes additional geometric or physical attributes, producing expressive volumetric representations within a compact and low-cost apparatus[cite: 2].

By combining established acoustic levitation techniques with a computational field model, this system offers a new class of volumetric visualization instrument: one that is inexpensive, safe, and capable of rendering dynamic geometric structures directly in free space[cite: 2]. The following sections describe the system architecture, levitation physics, rendering model, SEFI integration, control mechanisms, and feasibility considerations[cite: 2].

2 System Overview

The proposed volumetric rendering system uses acoustic levitation to position and manipulate a single illuminated particle as a dynamic voxel within an open three-

dimensional interaction volume[cite: 2]. The system is composed of four coordinated subsystems: an ultrasonic phased array, a levitated particle, an illumination module, and a geometric field engine that generates real-time voxel trajectories[cite: 2]. Together, these components form a compact, low-cost instrument capable of producing continuous volumetric curves and structures through persistence of vision[cite: 2].



Figure 1: Acoustic levitation hardware assembly featuring top and bottom ultrasonic transducer plates suspending a physical illuminated voxel particle in free space.

The ultrasonic phased array creates the acoustic field used for levitation and motion control[cite: 2]. By adjusting the phase relationships across its transducers, the array forms stable pressure nodes at precise spatial coordinates[cite: 2]. Dynamic modulation of these phase patterns enables smooth repositioning of the levitated particle throughout the interaction volume, allowing the voxel to trace arbitrary three-dimensional paths with high temporal coherence[cite: 2].

The levitated particle serves as the physical voxel[cite: 2]. A lightweight bead typically 0.5–2 mm in diameter is suspended within the acoustic field and illuminated externally[cite: 2]. Its motion, controlled by the phased array, forms the visible volumetric geometry[cite: 2]. Because the voxel is a tangible object rather than an optical artifact, it provides high contrast, clear visibility, and stable behavior under a wide range of environmental conditions[cite: 2].

The illumination subsystem enhances voxel visibility and encodes geometric or physical attributes[cite: 2]. A directed LED or laser diode illuminates the particle in synchronization with its position, ensuring crisp volumetric traces[cite: 2]. Color, intensity, or temporal modulation can represent curvature, field strength, or SEFI layer identity, adding expressive depth to the rendered structures[cite: 2].

The geometric field engine based on the Single Entity Field Interpretation (SEFI) generates the voxel’s trajectory[cite: 2]. SEFI worldlines, curvature fields, and layered field constructs are converted into coordinate streams that define the particle’s motion[cite: 2]. These coordinates are compiled into phase patterns for the ul-

trasonic array, enabling the voxel to physically trace the structure of the underlying field model in real time[cite: 2].

This integrated architecture provides a direct physical embodiment of geometric computation[cite: 2]. By combining established acoustic levitation techniques with a computational field model, the system renders dynamic three-dimensional structures in free space using a single controllable voxel[cite: 2]. The following sections detail the acoustic levitation physics, dynamic voxel rendering model, SEFI integration, control mechanisms, and feasibility considerations[cite: 2].

3 Acoustic Levitation Physics

Acoustic levitation uses ultrasonic pressure fields to suspend small particles at stable positions within an open volume[cite: 2]. When multiple transducers emit sound waves at the same frequency, their interference pattern forms regions of high and low acoustic pressure[cite: 2]. A particle placed within this field experiences a net force that drives it toward pressure minima or maxima, depending on its size and acoustic contrast[cite: 2]. By controlling the phase and amplitude of each transducer, these stable points known as acoustic nodes can be positioned precisely in three-dimensional space[cite: 2].

3.1 Standing Wave Formation

Ultrasonic transducers operating near 40 kHz generate sound waves with wavelengths on the order of 8–9 mm[cite: 2]. When arranged in a phased array, the superposition of these waves creates standing wave patterns[cite: 2]. The pressure distribution $p(\mathbf{r})$ at position $\mathbf{r}$ is determined by:

$$p(\mathbf{r}) = \sum_{i=1}^N A_i \cos(\mathbf{k}_i \cdot \mathbf{r} + \phi_i), \quad (1)$$

where A_i is the amplitude of transducer i , $\mathbf{k}_i$ is its wave vector, and ϕ_i is its phase offset[cite: 2]. Stable levitation points occur where the acoustic radiation force balances gravity and drag[cite: 2]. These points can be shaped and repositioned by adjusting the phase offsets ϕ_i , allowing the levitated particle to be moved smoothly through the interaction volume[cite: 2].

3.2 Particle Requirements

The levitated voxel is a lightweight bead whose acoustic and optical properties support stable suspension and clear visibility[cite: 2]. Suitable particles typically have:

- diameter: 0.5–2 mm[cite: 2],
- density: 20–200 kg/m³[cite: 2],

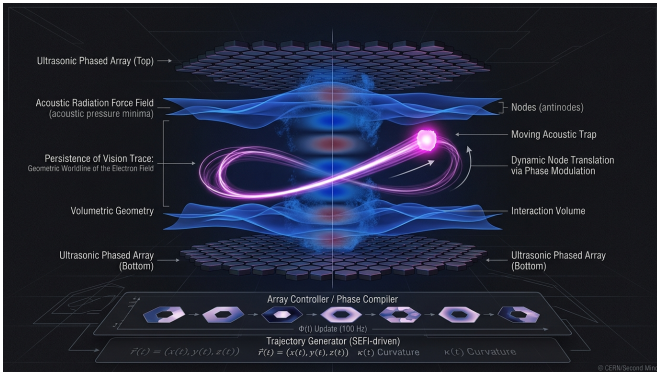


Figure 2: Schematic diagram of ultrasonic phased array pressure node translation driving a 3D parametric trajectory trace via persistence of vision.

- sufficient acoustic contrast to experience strong radiation forces[cite: 2],
- high optical reflectivity for crisp illumination[cite: 2].

Particles in this size range respond predictably to ultrasonic fields and remain stable under moderate environmental variation[cite: 2]. Their small mass allows rapid repositioning, enabling dynamic trajectories suitable for volumetric rendering[cite: 2].

3.3 Dynamic Positioning

Dynamic voxel motion is achieved by continuously updating the phase pattern across the array[cite: 2]. For a target coordinate $\mathbf{r}_t(t)$, the controller computes the required phase offsets:

$$\phi_i(t) = -\mathbf{k}_i \cdot \mathbf{r}_t(t). \quad (2)$$

This produces a moving pressure node that guides the particle along the desired trajectory[cite: 2]. Smooth motion requires phase updates at 100–500 Hz, stable amplitude control, compensation for environmental drift, and avoidance of destructive interference patterns[cite: 2]. When executed correctly, the particle can trace complex three-dimensional paths with high temporal coherence[cite: 2].

3.4 Stability Constraints

The stability of the levitated voxel depends on transducer alignment, phase accuracy, environmental noise, airflow, particle mass, and shape[cite: 2]. Within typical operating conditions, the system maintains stable levitation across a volume of several cubic centimeters, sufficient for rendering worldlines, curvature fields, and layered SEFI constructs using dynamic voxel motion[cite: 2].

4 Dynamic Voxel Rendering

Dynamic voxel rendering is achieved by moving a single acoustically levitated particle along a controlled three-dimensional trajectory and illuminating it in synchronization with its position[cite: 2]. When the particle moves rapidly enough, the human visual system integrates its motion into continuous volumetric curves and structures[cite: 2]. This persistence-of-vision effect allows a single physical voxel to represent complex geometric forms without requiring multiple particles or a dense volumetric medium[cite: 2].

4.1 Single-Voxel Volumetric Tracing

The levitated particle acts as a point emitter or reflector whose position is updated continuously by the ultrasonic phased array[cite: 2]. For a trajectory defined by:

$$\mathbf{r}(t) = (x(t), y(t), z(t)), \quad (3)$$

the voxel traces a visible curve in space when illuminated appropriately[cite: 2]. The apparent geometry is determined by the spatial path of the particle, the speed of traversal, the illumination timing, and the viewer’s persistence of vision[cite: 2]. Smooth, high-frequency motion produces crisp volumetric traces, while slower motion emphasizes discrete point positions[cite: 2].



Figure 3: Close-up laboratory visualization of a single 0.5–2 mm illuminated particle tracing a continuous light trajectory under high-speed acoustic modulation.

4.2 Illumination Model

Illumination is synchronized with the particle’s position to ensure that only the intended portions of the trajectory are visible[cite: 2]. A directed LED or laser diode provides a narrow beam that highlights the voxel with minimal spillover[cite: 2]. The illumination function $I(t)$ may encode geometric or physical attributes such as curvature, field strength, layer identity, entity membership, or temporal phase[cite: 2]. For example, curvature magnitude $\kappa(t)$ can be mapped to brightness:

$$I(t) = f(\kappa(t)). \quad (4)$$

4.3 Temporal Multiplexing

Although the system uses a single physical voxel, it can represent multi-layer or multi-entity structures through temporal multiplexing[cite: 2]. By cycling through multiple trajectories in rapid succession, the voxel renders layered SEFI constructs, multi-entity worldlines, stacked geometric fields, or composite volumetric shapes[cite: 2]. Let $\mathbf{r}_i(t)$ denote the trajectory for layer or entity i ; the system renders all layers by switching between trajectories at a frequency high enough to maintain perceptual continuity[cite: 2].

4.4 Volumetric Curve Quality

The clarity of rendered curves depends on voxel speed, illumination timing, acoustic stability, environmental conditions, and trajectory smoothness[cite: 2]. Optimized trajectories minimize abrupt accelerations that could destabilize levitation[cite: 2]. Illumination gating ensures that only valid positions contribute to the visible geometry, producing crisp, stable volumetric traces suitable for scientific visualization and geometric field exploration[cite: 2].

5 SEFI Geometry Integration

The volumetric rendering system derives its expressive capability from the integration of the Single Entity Field Interpretation (SEFI) with dynamic voxel motion[cite: 2]. SEFI provides a geometric field model that defines worldlines, curvature distributions, and layered field constructs[cite: 2]. These geometric outputs are translated directly into real-time trajectories for the levitated voxel, enabling the system to physically trace the structure of the underlying field model in free space[cite: 2].

5.1 Worldline Mapping

SEFI worldlines represent the geometric evolution of an entity’s position or state within the field[cite: 2]. Each worldline is defined as a parametric curve:

$$\mathbf{r}(t) = (x(t), y(t), z(t)), \quad (5)$$

which the voxel traces through dynamic acoustic positioning[cite: 2]. Because the voxel is a physical particle, the rendered worldline becomes a tangible, observable curve suspended in space[cite: 2].

5.2 Curvature Visualization

Curvature of a worldline is given by:

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}. \quad (6)$$

Curvature can be encoded visually through illumination intensity, color modulation, trajectory speed, or temporal gating, allowing the viewer to perceive geometric field behavior directly through the voxel’s motion and appearance[cite: 2].

5.3 Layered Field Constructs

SEFI defines multiple geometric layers that describe different aspects of field behavior[cite: 2]. These layers can be rendered using temporal multiplexing, where the voxel cycles through multiple trajectories at high frequency[cite: 2]. Layer identity can be encoded through color, brightness, traversal order, or modulation frequency, enabling visualization of SEFI’s multi-layer field constructs[cite: 2].

5.4 Entity and Interaction Representation

Multiple SEFI entities can be represented by cycling through their worldlines or by rendering interaction paths that describe relationships between entities[cite: 2]. Interaction geometry such as convergence, divergence, or coupling can be visualized through shared trajectory segments, synchronized illumination patterns, alternating voxel motion, or curvature-based emphasis[cite: 2].

5.5 Observer and Operator Geometry

SEFI includes geometric constructs that describe the observer’s frame or operator perspective[cite: 2]. These can be rendered as reference curves or volumetric scaffolds that contextualize entity motion[cite: 2]. The voxel can trace coordinate axes, reference frames, operator paths, or geometric anchors[cite: 2].

6 Control Architecture

The control architecture coordinates the geometric field engine, ultrasonic phased array, and illumination subsystem to produce stable, expressive volumetric rendering[cite: 2]. It is designed as a modular pipeline in which each stage transforms geometric information into physical behavior[cite: 2].

6.1 SEFI Engine to Trajectory Generator

The SEFI engine computes worldlines, curvature fields, and layered field constructs, producing $\mathbf{r}(t) = (x(t), y(t), z(t))$ [cite: 2]. The trajectory generator smooths discontinuities, enforces acceleration limits, resamples coordinates at a fixed control frequency, and prepares multiplexed trajectories[cite: 2].

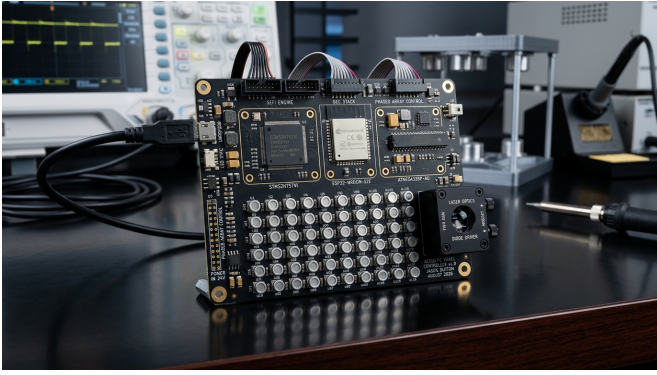


Figure 4: Custom embedded transducer driver board and laser diode optics assembly configured for high-frequency phase compilation and feedback control.

6.2 Trajectory to Phase Pattern Compiler

The phase pattern compiler converts each target coordinate into phase offsets:

$$\phi_i(t) = -\mathbf{k}_i \cdot \mathbf{r}(t). \quad (7)$$

This is performed at high frequency to maintain smooth voxel motion, with amplitude normalization and stability constraints[cite: 2].

6.3 Ultrasonic Array Controller

The array controller drives each transducer with the specified phase and amplitude, forming the pressure node that levitates the particle[cite: 2]. It provides synchronized timing, drift compensation, and safety monitoring[cite: 2].

6.4 Illumination Synchronization

The illumination subsystem receives the same coordinate stream and activates the light source only when the voxel occupies the correct spatial position[cite: 2]. Illumination parameters encode SEFI attributes such as curvature magnitude, layer identity, interaction strength, or temporal phase[cite: 2].

6.5 Feedback and Stability Monitoring

Optional feedback mechanisms include optical tracking of voxel position, acoustic field monitoring, environmental sensing, and adaptive phase correction[cite: 2].

6.6 Integrated Control Loop

1. SEFI engine generates geometric coordinates[cite: 2].
2. Trajectory generator refines and multiplexes paths[cite: 2].

3. Phase compiler converts coordinates into acoustic phase patterns[cite: 2].
4. Array controller forms and moves the levitation node[cite: 2].
5. Illumination controller lights the voxel in sync with its motion[cite: 2].
6. Feedback systems maintain stability and compensate for drift[cite: 2].

7 Safety Envelope

The volumetric rendering system operates within a well-defined safety envelope that ensures reliable performance and safe interaction[cite: 2]. Acoustic levitation at ultrasonic frequencies is non-ionizing and non-thermal, and illumination remains within eye-safe limits[cite: 2].

7.1 Acoustic Safety

The array operates near 40 kHz, above human auditory perception[cite: 2]. Acoustic output is limited to safe pressure levels, with monitoring of transducer load and thermal behavior[cite: 2].

7.2 Optical Safety

Directed LEDs or low-power laser diodes are used with controlled intensity, beam divergence, and gating, keeping illumination within safe exposure thresholds[cite: 2].

7.3 Particle Safety and Containment

The levitated voxel is a lightweight bead with low mass[cite: 2]. Particle size and material are chosen to avoid inhalation risk and toxicity, with optional containment structures for laboratory environments[cite: 2].

7.4 Environmental Stability

Environmental factors such as airflow and temperature gradients are mitigated through sensing, shielding, and stable mounting[cite: 2].

7.5 Operational Safeguards

Safeguards include thermal monitoring, automatic shutdown on unsafe acoustic output, illumination gating, trajectory validation, and controller watchdogs[cite: 2].

8 Feasibility Analysis

The system is low-cost, technically accessible, and based on established techniques[cite: 2].

8.1 Hardware Cost

Key components:

- Ultrasonic phased array: \$80–\$300[cite: 2]
- Microcontroller: \$10–\$40[cite: 2]
- Illumination subsystem: \$5–\$20[cite: 2]
- Levitated particles: negligible cost[cite: 2]
- Mechanical frame: \$10–\$30[cite: 2]

Total system cost: approximately \$200–\$400[cite: 2].

8.2 Engineering Feasibility

All core technologies—acoustic levitation, dynamic positioning, persistence-of-vision rendering, synchronized illumination—are established and documented[cite: 2].

8.3 Software and Control Complexity

The control architecture is modular and implementable using standard embedded systems and computational geometry techniques[cite: 2].

8.4 Technical Risks and Mitigations

Risks such as levitation stability, trajectory smoothness, environmental sensitivity, and illumination precision are manageable through engineering controls[cite: 2].

8.5 Development Path

1. Static levitation[cite: 2]
2. Slow dynamic positioning[cite: 2]
3. Full trajectory tracing[cite: 2]
4. Illumination synchronization[cite: 2]
5. SEFI-driven rendering[cite: 2]
6. Layered field visualization[cite: 2]

9 Applications for Acoustic & Spatial Sound Systems

The integration of SEFI-driven trajectory engines with acoustic levitation unlocks direct applications for audio engineering, transducer research, and spatial acoustic visualizer systems[cite: 2].



Figure 5: Interactive spatial audio application showing mid-air tactile particle manipulation and free-space visual anchor integration in an acoustic laboratory environment.

9.1 Acoustic Field Visualization & Transducer Calibration

Serves as a physical embodiment of SEFI, rendering acoustic energy distributions, wave interference patterns, curvature, and multi-layered field structures in mid-air[cite: 2]. It provides an experimental benchmark for testing phase manipulation, standing wave node stability, transducer phase control, and non-linear acoustic radiation forces[cite: 2].

9.2 Mixed Reality and Spatial Audio Interaction

Acts as a physical free-space cursor or spatial sound anchor, combining physical visual cues with directional audio transducers for spatial audio UX paradigms[cite: 2].

10 Conclusion

This paper presents a volumetric rendering system that uses acoustic levitation to position and manipulate a single illuminated particle as a dynamic voxel[cite: 2]. By integrating established ultrasonic control techniques with a geometric field model, the system produces expressive three-dimensional structures through persistence of vision[cite: 2]. The Single Entity Field Interpretation (SEFI) provides the mathematical foundation for generating worldlines, curvature fields, and layered constructs, which are translated directly into real-time voxel trajectories[cite: 2].

The system’s architecture is compact, low-cost, and grounded in well-validated physical principles[cite: 2]. Its modular control pipeline enables stable and precise voxel motion, and the resulting volumetric traces offer a clear method for visualizing geometric and acoustic phenomena[cite: 2]. By demonstrating that complex geometric structures can be rendered using a single levitated voxel, the system establishes a foundation for future volumetric display and spatial acoustic technologies[cite: 2].